Abanisenioluwa Kolawole Orojo

Waco, Texas · akorojo@labackdoor.com · labackdoor.com
linkedin.com/in/abaniseorojo · github.com/AKOrojo · Google Scholar · ORCID 0009-0005-9448-1929 · DBLP

RESEARCH PROFILE

Computer Science PhD candidate at Baylor University working at the intersection of AI and systems security. I build and evaluate AI agents that find and fix vulnerabilities on live systems, and the cyber ranges, benchmarks and detection tooling around them. Creator of SysRepair-Bench (313 reproducible attack-and-repair scenarios), lead developer of an asynchronous red/blue self-play RL pipeline on a 32-GPU, three-site cluster, and author of twelve papers and preprints. Four years of hands-on detection and response (SIEM/IDS monitoring, phishing-triage automation, blue-team defense under live red-team pressure). Fluent in Python, C++, Rust, Java, Go, SQL and Bash. Also a graduate research fellow at BASE Research, studying how the AI safety field frames risk and whose voices it leaves out. Seeking Summer 2027 internships in detection and response and AI security.

EDUCATION

Baylor University — Waco, TX
Doctor of Philosophy in Computer Science (Advisor: Dr. Erika A. Leal) — *Jan 2023 – Expected Dec 2027*

- Research: autonomous vulnerability remediation, AI-driven threat detection, network-protocol security, distributed access control, cyber-physical systems security.
- Affiliations: Central Texas Cyber Range (CTCR); Cybearsecurity (la backdoor) Lab; Baylor Multimedia Lab.

Webster University — St. Louis, MO
Master of Science in Cybersecurity — *Aug 2022*

- Thesis: *Navigating the Digital Maze: Exploring the Intersection of Social Media, Privacy, and Security.*

Webster University — St. Louis, MO
Bachelor of Science in Computer Science, Minor in Management — *May 2020*

- Valedictorian, undergraduate class of 2020.

RESEARCH EXPERIENCE

Baylor University, Central Texas Cyber Range — Waco, TX
Doctoral Researcher and Cybersecurity Research Assistant — *Jan 2023 – Present*

- Lead developer of the three systems below; own design, implementation and evaluation end to end. Collaborate across a five-person, three-country research team (Baylor, Ghana, Bangladesh) and with NCSA (Illinois) and Google Cloud for compute.
- Build and operate the range's research infrastructure on a six-node VMware vSphere cluster (312 cores, 2.5 TB RAM, 106 TB storage, 12 NVIDIA L40 GPUs, 74 isolated subnets): the malware-testing harness in use today and Terraform/Ansible provisioning for 30+ virtualized attack-and-defense lab scenarios used for red/blue exercises and incident-response training.
- Designed MAADER, a cyber-physical testbed combining emulation, simulation and physical components to study attacks on critical infrastructure.
- Red-team large language models and author evaluation material for security-focused LLMs.

SysRepair: Evaluation Platform for Autonomous System Remediation — *2025 – Present*

- Built and open-sourced 313 reproducible Docker/VM scenarios in six suites (137 modern CVEs including Log4Shell, regreSSHion, PwnKit, Dirty Pipe and the XZ backdoor; an Active Directory lab covering Zerologon, Kerberoasting, DCSync, ADCS ESC1–8 and PrintNightmare; Windows containers; kernel-pinned VMs; free-roam hosts with planted persistence). Each scenario is graded by a deterministic verifier that re-runs the original exploit and requires the service to still serve.
- Designed a uniform solver API on Inspect AI so six agent architectures (ReAct, Reflexion, Plan-and-Solve, LATS, Agentless, NeuroPlan) run unchanged against one sandbox; the harness runs 15 concurrent containers per host, Hyper-V/KVM VMs and local vLLM on L40 GPUs, and logs tokens, wall-clock time and cost per success.
- Created NeuroPlan, a neural-symbolic solver that lifts a live host (osquery) into PDDL and executes a precondition-checked plan: 76% end-to-end remediation with every plan reviewable before execution. The best LLM solver reaches pass@5 of 65.4%; Plan-and-Solve uses 38× fewer tokens per success than ReAct. Agents are weakest on compensating controls (44.2% success@5).

Asynchronous Red/Blue Self-Play RL on a Multi-Site GPU Cluster *2026 – Present*

- Building an asynchronous RL post-training system for attacker/defender self-play: decoupled rollout and trainer workers, partial rollout with buffered resumption and KV-cache rebuild after each weight update, and importance-ratio-corrected group-relative advantages with dynamic sampling to bound policy staleness.
- Scaled training to 32 GPUs across three sites (NCSA Delta, local GPU servers, Google Cloud) over NCCL; adapted and fixed collective-communication configuration for heterogeneous cross-site interconnects.

LLMs for Network Defense and Threat Detection (TrAice, ByteFlow, MRNN, AW-TRBAC) *2023 – 2025*

- Engineered TrAice, a distributed LLM pipeline in Python that ingests high-volume PCAP and host/network logs and produces analyst-ready summaries; developed ByteFlow, a byte-level LLM for deep packet inspection (ACSAC Workshops 2025).
- Built a Multi-Recurrent Neural Network that forecasts software-vulnerability trends, beating ARIMA and LSTM baselines by 32.78% (NLPAICS 2024).
- Architected AW-TRBAC, a time-and-role-based access-control framework for distributed NoSQL databases with node-identity management, and paired it with semantic variational autoencoders for dynamic threat detection (SAM 2025, ICAI 2025).
- Led data collection and modeling over a 250,000-message Telegram corpus to surface reliable firsthand accounts from conflict zones (ASONAM 2024; NSF Grant No. 2245983).
- Built client-side video analysis for content moderation over Media-over-QUIC (ACM Mile-High Video 2026) and an evaluation of ABR switching for time-shifted MoQ clients (arXiv 2026).

BASE Research San Francisco Bay Area
Graduate Research Fellow (part-time, remote) *Oct 2026 – Present*

- Use computational text analysis to examine how the AI safety field defines and prioritizes risk, whose perspectives shape the discussion, and which voices and countries are underrepresented; compare catastrophic-risk framings (including loss of control) with present-day concerns such as bias, misuse and reliability.
- Building a corpus of academic papers, research-community discussions, safety-organization publications and summit documents, tracking sources, author affiliations and geographic representation. Analysis combines embeddings, topic modeling and LLM-assisted classification checked against human-coded examples, to study changes in risk framing over time and engagement between communities through citation patterns.

Micro1 Remote
Cybersecurity AI Trainer (part-time, remote) *2025 – 2026*

- Authored training data and evaluation benchmarks for security-focused LLMs: exploit analysis, alert triage, incident-response playbooks and adversary-behavior characterization.

INDUSTRY EXPERIENCE

Apex Ezone United Kingdom (Remote)
Security Engineer Intern, Backend Systems *Aug 2021 – Dec 2022*

- Monitored and investigated host and network security events with IDS/SIEM tooling and Splunk log analysis for a high-traffic eSports platform; drove incidents from alert to resolution with a remote UK engineering team.
- Automated phishing triage and investigation with open-source tooling, cutting phishing incidents 40% and incident-response time 30%; ran Nessus/Nmap vulnerability assessments and hardened TLS/SSL configurations.
- Designed and secured REST APIs and containerized Django microservices backed by PostgreSQL and Redis.

Leti Arts Greater Accra, Ghana
Software Developer Intern *May 2018 – Aug 2018*

- Shipped 100+ mini-games in C++ and in-house web applications in Angular and Node.js (user engagement +30%); built SQL/Hadoop/Spark analytics tools; introduced Docker deployment (setup time −50%) and Git-based review (bug reports −70%).

PUBLICATIONS

[*] *first author. Full record on Google Scholar, DBLP and labackdoor.com/publications.*

Under review and preprints

[P1] * **A. Orojo**, et al. Typed, Calibrated, and Manipulable: Calibration Collapse and Adversarial State Injection in Non-Autoregressive Decision Models for Intrusion Detection and Alert Triage. Under review, ACM CODASPY 2027.

[P2] * **A. Orojo**, W. Elumelu, E. El-Mahmoud, E. Leal. SysRepair-Bench: A Benchmark for AI Agents' Ability to Remediate Real-World System Vulnerabilities. Under review, ICLR 2027.

[P3] * **A. Orojo**, W. Elumelu, E. El-Mahmoud, E. Leal. SysRepair: A Benchmark and Multi-Architecture Approach to Autonomous Vulnerability Remediation. Under review, ACM CODASPY 2027.

[P4] * **A. Orojo**, W. Elumelu, E. El-Mahmoud, E. Leal. Autonomous System Vulnerability Remediation: A Survey of Agentic AI, Reinforcement Learning, Benchmarks, and Operational Safety. Under review; preprint on Research Square, 2026.

[P5] * **A. Orojo**, T. Redoy, S. Afzal, A. C. Freeman. An Evaluation of ABR Switching for Time-Shifted Clients in MoQ. arXiv:2606.26368, 2026.

Peer reviewed

[C1] A. Freeman, J. Bellert, **A. Orojo**, G. Speegle. Client-Side Video Analysis for Content Moderation with MoQ. Proc. 5th ACM Mile-High Video Conference (MHV), pp. 93–99, 2026.

[C2] * **A. Orojo**, E. El-Mahmoud, E. Leal, P. Rivas. ByteFlow: A Byte-Level LLM for Deep Packet Inspection and Network Intelligence. WAITI Workshop, ACSAC 2025 (IEEE), pp. 357–365. DOI 10.1109/ACSACW69556.2025.00046.

[C3] **A. Orojo**, E. El-Mahmoud, S. Hutton, M. Donahoo, W. Elumelu, et al. A Unified Framework Incorporating AW-TRBAC and Semantic Variational Autoencoders for Dynamic Threat Detection and Access Control. ICAI 2025 (Springer CCIS, 2026).

[C4] * **A. Orojo**, B. Khanal, E. El-Mahmoud, J. Yu, M. B. Rashid, et al. Baylor Environmental AI Research System (BEARS): An Agentic AI Project to Combat Climate Change. ICAI 2025 (Springer CCIS, 2026).

[C5] * **A. Orojo**, E. El-Mahmoud, G. Speegle. Assessing the Impact of Access Control Policies on Data Accessibility in Distributed NoSQL Environments. SAM 2025 (Springer CCIS, 2026).

[C6] * **A. K. Orojo**, W. C. Elumelu, O. O. Orojo. Predicting Software Vulnerability Trends with Multi-Recurrent Neural Networks: A Time Series Forecasting Approach. NLPAICS 2024 (ACL), pp. 42–47.

[C7] * **A. Orojo**, P. Bhagat, J. Wilburn, M. J. Donahoo, N. Vishwamitra. Leveraging Secure Social Media Crowdsourcing for Gathering Firsthand Account in Conflict Zones. IEEE/ACM ASONAM 2024, Springer LNCS 15212, pp. 160–170. DOI 10.1007/978-3-031-78538-2_14.

PRESENTATIONS AND TALKS

- **From Measurement to Co-Adaptive Defense: Autonomous Vulnerability Remediation on Live Systems.** Texas Security Workshop 2026, UT Dallas, Richardson, TX, Sept. 26, 2026. Research talk (one of twelve selected presentations). Slides: labackdoor.com/talks.
- **Kyere: Watching isn't knowing.** Finalist pitch and live demo (team Barney and Friends), WaCode 2026 Hackathon, Baylor University, 2026. Placed 2nd.

HONORS AND AWARDS

- **2nd place**, WaCode 2026 Hackathon, Baylor University (Kyere, a study app where students prove understanding by teaching an AI student).
- **2nd place** (2025) and **3rd place** (2024), Southwest Collegiate Cyber Defense Competition regionals, Baylor University.
- **Top 30 nationally** (about 475 teams), National Cyber League, 2020–2024.
- Three Baylor Graduate School travel awards.
- Undergraduate Valedictorian, Webster University, 2020.

TEACHING, MENTORSHIP AND LEADERSHIP

Competition Coach and former Network Security Lead, CCDC — Waco, TX
Baylor University — *2023 – Present*

- Network security lead (2023–2025) for live blue-team defense against a professional red team (hardening Linux/Windows hosts, alert triage, persistence hunting, incident reports); now coach the team on detection, hardening and response playbooks.

Director, Cybersecurity Summer Camp — Waco, TX
Central Texas Cyber Range, Baylor University — *Summer 2023, 2025*

- Developed and directed the training curriculum; participants' threat-awareness and prevention understanding rose 90%.

Teaching Assistant, Data Communications Waco, TX
Baylor University *Jan – May 2023*

- Mentored 30+ students in TCP/IP networking, routing protocols, DNS and data-communications principles.

OPEN-SOURCE RESEARCH SOFTWARE

- **SysRepair-Bench** (Python, Inspect AI): 313-scenario benchmark for AI agents that remediate live-system vulnerabilities. github.com/BaylorSecurityLab/sysrepair-bench
- **Autopatch** (Python, LangGraph): six-agent pipeline that discovers, remediates and verifies infrastructure vulnerabilities behind a policy-gated, signed execution sandbox wired to OpenVAS, Trivy, Nuclei and NVD/EPSS/CISA-KEV feeds. github.com/AKOrojo/autopatch
- **Stackforge** (Rust, Python): modular networking stack and automata framework with native Python bindings.
- **DDoS attack simulation framework** (Terraform, Ansible): infrastructure-as-code reproduction of adversary techniques for validating detection and mitigation.
- **Secure crowdsourcing dataset**: 250,000-message corpus behind the ASONAM 2024 paper.

TECHNICAL SKILLS

Languages	Python, C/C++, Rust, Go, Java, SQL, Bash, JavaScript/TypeScript, PHP
AI and ML	PyTorch, TensorFlow, scikit-learn, vLLM, Inspect AI, LangGraph, asynchronous RL post-training, NCCL, LLM agents and red-teaming, time-series forecasting, PDDL / Fast Downward
Detection and Response	Splunk, IDS/IPS, PCAP/Wireshark, osquery, auditd, OpenVAS, Nessus, Nmap, Metasploit, Trivy, Nuclei, MITRE ATT&CK / D3FEND, CVSS and CISA KEV triage, adversary emulation
Systems and Infra	Linux internals (PAM, sudoers, cron, SUID), Windows Server and Active Directory, Docker, Kubernetes, VMware vSphere, Proxmox, KVM/Hyper-V, Terraform, Ansible, CI/CD, GCP, Azure
Networking	TCP/IP, DNS, BGP, TLS/PKI, QUIC/Media-over-QUIC, NDN, ns-3, deep packet inspection
Data	PostgreSQL, Redis, MongoDB, Hadoop, Spark
Hardware security	Microarchitectural attacks (cache and branch-predictor side channels, Flush+Reload), PMU-based detection, gem5/WeMu emulation

CERTIFICATIONS

Offensive Security Certified Professional (OSCP) · CompTIA CySA+ · SOC Analyst Level 2 · MITRE ATT&CK (RangeForce) · LPI Linux Essentials

SERVICE AND PROFILES

Lab and portfolio: labackdoor.com. References available on request.